

1

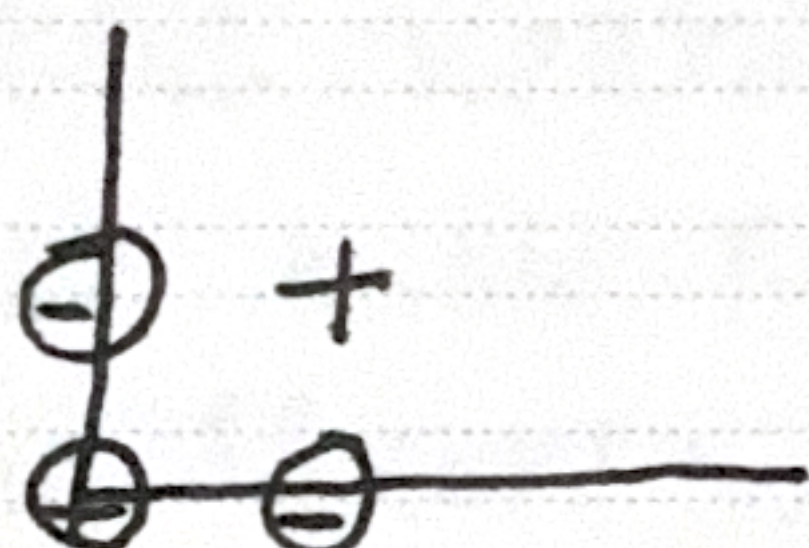
1

A) It's a line along the shadow of where 3D sigmoid is 0.5.

B) Yes. $\text{iter} = 2000$ and $\text{lr} = 0.1$

2

A)



No. the data is not linearly separable.

B) The smallest has 3 hidden units and 1 final unit because you need 4 linear separators.

3

A) Not with 2 hidden units.
with 3 you can. But with 2 it
get's stuck in local minima.

B) No. There is one data point
that is very fishy. (Noisy possibly)

2

A) 1 output neuron. Linear activation.
We need both positive and negative
integers. Squared loss.

B) Sigmoid, we need 1 single probability.
NLL.

C) Softmax, we need multiclass
prob, NLLM.

D) n -topics output neurons. ^{Step (Sigmoid)}
 $\sum NLL$.